MIL-S-51201A(MU)
13 January 1969
SUPERSEDING
MIL-S-51201(MU)
14 October 1964

MILITARY SPECIFICATION

SAMPLING AND ANALYZING KIT, CBR AGENTS, M19

1. SCOPE

1.1 This specification covers one type of kit for use in the sampling and analyzing of CBR agents.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Steel Strapping, Flat.
TT-W-572 - Wood-Preservative; Water Repellent.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Box, Fiberboard.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.

MILITARY

MIL-D-16791 - Detergents, General Purpose (Liquid, Nonionic).
MIL-C-46842 - Cushioning Material, Unicellular Polyethylene Foam (For Packaging Purposes).

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

FSC 6665

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DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

D150-2-1 - Marking Diagram for Shipping Containers.
DL5-77-353 - Sampling and Analyzing Kit, CBR Agents, M19.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials, cited on Drawing DL5-77-353 or on the subsidiary drawings, shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the M19 kit shall conform to the specifications and drawings listed on Drawing DL5-77-353 and subsidiary drawings.

3.1.2.1 Lamp assembly. The lamp assembly, conforming to Drawing B5-77-472, shall not leak when tested as specified in 4.2.4.1.

3.1.2.2 Bottle assembly. The screw cap bottles containing water, shall show a loss of water of no greater than 0.2 percent by weight and no evidence of water shall be observed in the heat sealed barrier bags when tested as specified in 4.2.4.5.

3.2 Manufacture and assembly. The M19 kit shall be assembled as shown on Drawing E5-77-353-1 or E5-77-353-2 as specified in the contract or purchase order (see 6.2).

3.2.1 Stitching. All rows of stitching shall be straight or regular lines free from loose stitches or thread loops. When stitching is not of continuous thread, each thread or break shall be backstitched, whenever practicable, for no less than two inches. All seams and hems shall be backstitched at the ends with two or three stitches to prevent raveling.

3.2.2 Clip. When attaching clips, the duck shall be inserted the full depth of the clip. The pressure used in assembling the clip to the duck shall be such as not to cut, break, or weaken the duck. The clip shall form a smooth, flat, tight, and compact assembly at the end of the duck.

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3.3 Paint fluorescence. The markings on the individual components shall be fluorescent and legible when irradiated with the ultra violet source lamp when tested as specified in 4.2.4.2.

3.4 Contamination of filter paper. The filter paper shall be free from grease, oil, hand lotions, or other foreign matter when tested as specified in 4.2.4.3.

3.5 Pump capacity. The pump shall have a capacity of 4.0 ± 0.3 liters of air per minute at 2.0 ± 0.1 inch of mercury suction when crank is rotated at 60 ± 5 rpm (see 4.2.4.4).

3.6 Workmanship. The kit shall be free from damage (dents, breakage, scratches, punctures, or tears) and contamination (grease, oil, or foreign matter).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the kits assembled by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of detector tubes, tickets, crayons, or reagent chemicals shall be represented in any one lot of assembled kits.

4.2.2 Sampling. Each kit in the lot shall be examined and tested. Ten bottle assemblies shall be selected at random for the leakage test.

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4.2.3 Inspection procedure. Each kit shall be examined in accordance with the list of defects. The sample bottle assemblies shall meet the leakage requirements of 3.1.2.2. The acceptance number for the bottle assemblies is 0.

4.2.3.1 List of defects.

(a) Sampling and analyzing kit, CBR agents, M19. (Dwg E5-77-353.)

1. Component missing.
2. Component incorrectly assembled.
3. Marking missing.
4. Stitches loose or looped.
5. Broken stitches not backstitched.
6. Workmanship (3.6).
7. Protective finish missing.
8. Paint fluorescence (4.2.4.2).
9. Filter paper contaminated (4.2.4.3).
10. Paint on threads of pressure relief valve.

(b) Pump, vacuum (Dwg D5-77-394).

1. Protective finish missing.
2. Component incorrectly assembled.
3. Pump capacity (4.2.4.4).
4. Workmanship (3.6).

(c) Extractor, smoke (Dwg C5-77-382).

1. Component incorrect or incorrectly assembled.
2. Hole missing or incorrectly located.
3. Protective finish missing.
4. Upper and lower clamp surfaces not mating properly.
5. Item nonfunctioning.
6. Workmanship (3.6).

(d) Burner, alcohol (Dwg C5-77-468).

1. Component missing or incorrectly assembled.
2. Protective finish missing.
3. Lamp assembly leakage (4.2.4.1).
4. Workmanship (3.6).

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(e) Vapor sampling kit (Dwg D5-77-452).

1. Component missing or incorrectly assembled.
2. Hardware nonfunctioning.
3. Stitches loose or looped.
4. Broken stitches not backstitched.
5. Marking missing, incorrect, or illegible.
6. Protective finish on the sampling head missing.
7. Workmanship (3.6).

4.2.4 Tests. Tests shall be conducted as follows:

4.2.4.1 Lamp assembly leakage. Fill the lamp assembly with a solution containing 4 grams of water-soluble detergent, conforming to type I of MIL-D-16791, per one gallon of water and allow the lamp to stand for one hour plus 15 minutes and then examine for leakage. Clean and dry the lamp before reassembly.

4.2.4.2 Paint fluorescence. Place the kit in a dark room and subject the components to the ultraviolet light source (long wave, 3660 Angstroms) held within 8 to 12 inches from the markings on the components.

4.2.4.3 Filter paper contamination.

(a) Preparation of specimen. Take 4 specimens from each roll of filter paper as follows: Unroll 12 inches of filter paper and cut 2 one-inch specimens. Unroll 24 additional inches and cut 2 one-inch specimens. Acceptance of roll shall not be placed on length in feet.

(b) Procedure.

(1) Hold a specimen with teflon tip forceps and add one drop of a solution containing 0.001 gram (g) of di-p-biphenylthiocarbazone (DBT) in 2 milliliters of acetone. A color of gray-green indicates no contamination present. Any color of tan, black, red, pink or blue which persists after 90 seconds indicates contamination present.

(2) Place a one square centimeter piece of the specimen in a small test tube, add a drop of a solution containing 0.1 g of DB3 (4-4-nitrobenzylpyridine) in 2 ml of acetone, add one drop of ammonium hydroxide. A white to yellow color indicates no contamination present. A peach-brown, blue-purple, or red color indicates contamination present.

4.2.4.4 Pump capacity. Place the pump in a suitable fixture and crank it at the required speed. Check the folume of air delivered.

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4.2.4.5 Bottle assembly leakage. Bottles heat sealed in barrier material as specified shall be subjected to a temperature of 65° ± 2°C for 24 hours prior to obtaining the initial weight. Bottles shall be aged for two weeks at 65° ± 2°C and checked for weight loss of water. Heat sealed barrier bags shall be torn open and observed for any water in the bag. Calculate the weight loss of water as follows:

$$\text{Percent weight loss of water} = \frac{\text{Initial weight - Final weight}}{100} \times 100$$

5. PREPARATION FOR DELIVERY

5.1 Packaging, level A. Each kit shall be packaged in a fiberboard box conforming to V3c, style FOL of PPP-B-636 and having inside dimensions of approximately 24 by 8-5/8 by 16 inches length, width and depth respectively. Each box shall be provided with two cushion pads, one in the bottom and one in the top. Pads shall have inside dimensions of approximately 23-3/4 by 8-1/2 by 1-1/2 inches length, width and depth respectively and be of material conforming to polyethylene foam, MIL-C-46842. The top pad shall be notched to provide a recess for the handle of the case. Filler pads of the same type material as the box shall be included, as necessary, to restrict free movement of the box contents. The box shall be closed and all seams and joints sealed with tape in accordance with the appendix to PPP-B-636.

5.2 Packing, level A. The kit, packaged in accordance with 5.1, shall be packed in either a nailed wood box conforming to style 4, class 2, PPP-B-621; cleated plywood box, overseas type, PPP-B-601 or wirebound style 2A or 3, class 3, PPP-B-585. Unless otherwise specified (see 6.2), the wood shipping container or the finished wood parts thereof shall be completely immersed for a minimum of three minutes in a water repellent wood preservative conforming to type II, composition A of TT-W-572. The box shall have inside dimensions of approximately 24-5/8 by 9 by 16-7/8 inches length, width and depth respectively. Fiberboard filler pads of material conforming to weather resistant class, PPP-F-320 shall be included as necessary to restrict free movement of the package. The box shall be closed and strapped in accordance with the appendix to the respective box specification. Strapping shall conform to class B, QQ-S-781.

5.3 Marking. In addition to any special marking required by the contract or order and the special marking specified herein, unit packages shall be marked in accordance with MIL-STD-129 and shipping containers shall be marked in accordance with D150-2-1. All markings on the shipping containers shall be clear, legible and permanent after the application of the wood preservative specified in paragraph 5.2.

5.3.1 Special marking. The earliest expiration date for any package in the shipping container shall be shown to the right of the "date packed" information as follows: "EXPIRES, Month/Year".

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6. NOTES

6.1 Intended use. The kit covered by this specification is intended for use in sampling, detecting, and identifying toxic chemical warfare agents and toxic smokes in the field, in the air, in the water, and on material surfaces and for conducting partial qualitative analysis of such materials. It is also intended for use in sampling BW and RW agents on surfaces only for further processing by appropriate laboratories.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Whether aluminum case (E5-77-353-1) or plastic case (E5-77-353-2) is required.
(c) Whether wood preservative is not required.

6.4 Component specifications. The following list of specifications gives the components of the Sampling and Analyzing Kit, CBR Agents, M19:

Specification titles	Number
Packet, Sodium Pyrophosphate Peroxide	MIL-P-50014
Tablet, Compound 34	MIL-T-51023
Detector Tubes, Chemical Agent	MIL-D-51082
Detector Ticket, Anticholinesterase Agent	MIL-D-51083
Substrate Solution and Dispenser	MIL-S-51189
Light, Illuminating and Ultraviolet	MIL-L-51196
Reagents, Solvents, and Test Papers for Sampling and Analyzing Kit, CBR Agents	MIL-R-51197
Repair Kit, CBR Agent Sampling and Analyzing Kit, M32	MIL-R-51198
Refill Kit, Analyzing Components, CBR Agent Sampling and Analyzing Kit, M33	MIL-R-51199
Sampling Kit, CBR Agent, M34	MIL-S-51200
Soil Sampling Kit	MIL-S-51210
Paper, Chemical Agent Detector, VGH, ABC-M8 Booklet Of	PD 197-54-838

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 6665-A235

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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